

Multimodal Geometric Deep Learning for Urban Commercial Potential Segmentation using Open Geospatial Data

Trien D. Nguyen^[0009-0005-8769-5363], Lam Mai^[0000-0002-8701-8961], Quynh X. N. Nguyen and
Nguyen M. H. Nguyen

The University of Danang - Vietnam-Korea University of Information and Communication
Technology, Danang, Vietnam
mlam@vku.udn.vn

Abstract. Data-driven site selection in the food and beverage (F&B) industry traditionally excludes independent operators due to a reliance on proprietary revenue data. To overcome this structural barrier, we propose a novel proxy-label framework for Coffee & Tea site potential classification, operating entirely on open-source geospatial signals including Google Maps community reviews, OpenStreetMap point-of-interest density, and VIIRS night-time light imagery. Our adapted multimodal geometric deep learning architecture processes $11 \times 11 \times 11$ neighborhood spatial tensors via a two-stage geometry refinement with spatial-channel attention mechanism, integrated with a Wide-Cross-Deep network for tabular features. Validated on a city-stratified held-out test set of 14,797 grid cells across three major Vietnamese cities, the model achieves a Macro F1 score of 0.7613, outperforming linear and neural baselines such as Logistic Regression, Wide & Deep, and Deep & Cross Networks. Furthermore, rigorous ablation studies confirm that the spatial attention mechanism is the most critical component for minority class discrimination, emphasizing the necessity of spatial context over flattened representations. Demonstrating a practical recall of 0.84 for high-potential locations, this framework proves that open data can successfully supervise complex models, providing an accessible decision-support tool for fragmented markets.

Keywords: Geometric Deep Learning, Urban Remote Sensing, Point-of-Interest Data, Spatial Attention, Spatial Supervision.

1 Introduction

Commercial site selection is vital for the long-term success of Food and Beverage (F&B) businesses, especially given the rapid expansion of Vietnam's highly competitive Coffee and Tea sector [1, 2]. While deep learning frameworks like DeepStore [3] and O²-SiteRec [4] offer data-driven alternatives by integrating spatial context with mobility signals [3, 5–7], these approaches depend heavily on proprietary transaction data, which are often unavailable in fragmented emerging markets such as Vietnam. In parallel, recent studies in open-data urban analytics have investigated individual open

geospatial signals, including OpenStreetMap (OSM) points-of-interest [8, 9], VIIRS Night-Time Light (NTL) imagery [10, 11], and user reviews [12]. Together, these two research directions highlight the value of urban geospatial data for commercial site analysis, but few studies combine multiple open data sources in a single framework for fine-grained site evaluation. To address this gap, we propose a multi-modal geospatial framework using entirely open-source data to estimate commercial potential for the data-scarce Coffee & Tea segment across Hanoi, Da Nang, and Ho Chi Minh City.

The main contributions of this study are: (1) A proxy labeling method generating three-level site potential scores by combining macroeconomic NTL data, local OSM POI networks, and machine-translated Google Maps reviews. (2) An adapted multi-modal Wide & Deep model for data-scarce settings that simultaneously processes 3D spatial data and tabular features. (3) A fine-grained urban benchmark dataset of 250×250 meters grid cells, offering a practical reference for Vietnam. To provide an overview of the complete pipeline, Fig 1. summarizes the proposed framework.

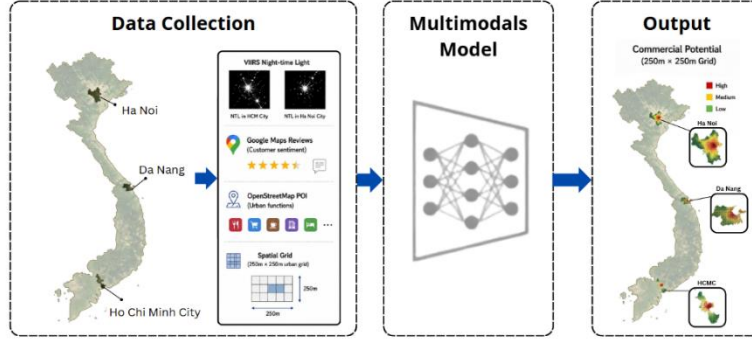


Fig. 1. Overview of the proposed multimodal geospatial framework for commercial site potential estimation using open-source urban data.

2 Methodology

2.1 Problem Formulation and System Overview

Following the overall pipeline illustrated in Fig 1., this section formalizes the proposed multimodal framework for commercial site potential estimation. The proposed framework operates entirely on open-source urban signals and consists of three stages: (1) multimodal data acquisition and spatial partitioning, (2) feature and proxy-label engineering, and (3) interaction-aware multimodal learning. We divide the target urban area U (Hanoi, Da Nang, Ho Chi Minh City) into a set of non-overlapping grid cells $G = \{g_1, g_2, \dots, g_N\}$. Following small-scale urban analysis practices, each cell g_i has a spatial resolution of 250×250 meters. For each grid cell g_i , we build a multimodal feature set $\mathcal{X}_i$, which includes tabular statistical vectors describing the cell's own local characteristics, along with 3D spatial data that captures information from its surrounding area. To define the ground truth for model training, each cell is assigned a discrete

label $Y_i \in 0,1,2$, corresponding to low, medium, and high commercial potential. These labels are determined based on a combined evaluation of existing store density and quality indicators within each cell, reflecting its actual market attractiveness. As a result, the research problem can be defined as learning an optimal mapping function $f: \mathcal{X}_i \rightarrow Y_i$, with the goal of maximizing prediction accuracy for a site's spatial potential using only open-source data. The detailed architecture of this mapping mechanism is illustrated in Fig 2.

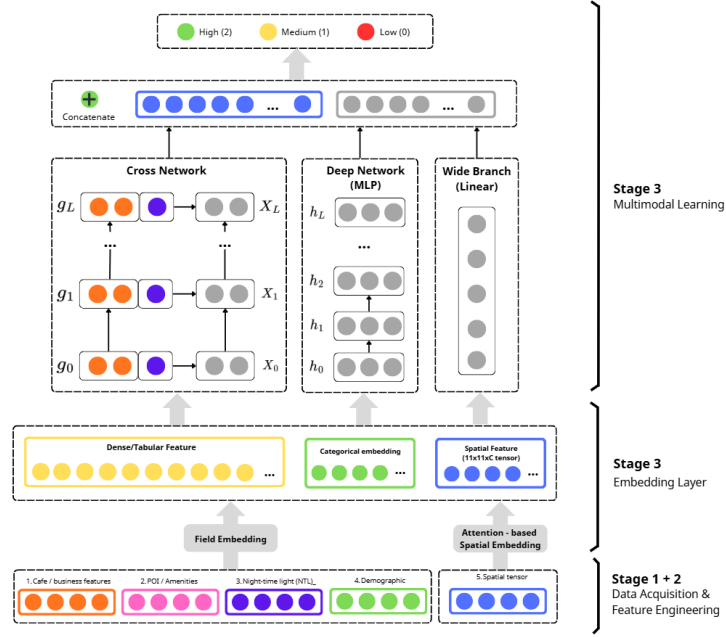


Fig. 2. Detailed architecture of the proposed multimodal learning model. Input features (Stage 1 & 2) are processed through a multimodal embedding layer before being passed to three parallel branches - Cross Network, Deep Network (MLP), and Wide Branch (Linear) - whose outputs are concatenated and passed through a dense layer before the final softmax classifier for three-class commercial potential prediction.

2.2 Multi-source Data Acquisition

To build a comprehensive multimodal dataset, satellite images, geospatial platforms, and user-generated content are combined to capture urban economic activity, functions, and human perception. Specifically, we extract annual median VIIRS DNB composites from 2023 via Google Earth Engine, using per-cell mean radiance as a macroscopic economic proxy. Micro-level functional characteristics are captured via Point-of-Interest (POI) data from OpenStreetMap (Overpass API) [8], classified into eight categories: Food, Commercial, Leisure, Transport, Office, Education, Residential, and Other. To capture both physical business density and subjective market performance, commercial

establishment metadata and User Generated Content (UGC) are extracted from Google Maps. This includes spatial coordinates, venue counts, and total review volumes, etc. alongside text reviews that are machine-translated and processed via VADER sentiment analysis to derive average ratings and customer satisfaction scores.

2.3 Feature Engineering

The continuous urban area is divided into a uniform grid network $\mathcal{G} = g_1, g_2, \dots, g_N$, where each cell g_i has a spatial size of $250m \times 250m$. To meet the dual-input design of the proposed deep learning model, the features of each cell g_i are organized into two different formats. First, a tabular vector $x_i \in R^D$ is created to represent the internal characteristics of the cell, comprising log-transformed POI category counts, POI Density, POI Entropy, NTL_log, and geographic centroid coordinates (D=13). Second, to capture the surrounding spatial context, a 3D tensor $\mathcal{T}_i \in R^{K \times K \times C}$ is constructed. This is done by taking a $K \times K$ window centered on g_i , which includes C feature channels from nearby cells. In this study, $K=11$ and $C=11$, resulting in an $11 \times 11 \times 11$ tensor per grid cell.

2.4 Label Engineering and Geometry Supervision

Since proprietary revenue data is unavailable, we design a heuristic Commercial Potential Score (Vraw) combining market size, customer flow, satisfaction, and neighborhood economic strength:

$$V_{raw} = \ln(1 + N_{cafe}) \times \ln(1 + N_{review}) \times Q_{score} \times E_{neighbor} \quad (1)$$

Here, N_{cafe} and N_{review} represent the number of Coffee & Tea businesses and the total number of reviews in the cell. A logarithmic transformation is applied to these values to reduce the effect of extreme cases, such as very dense tourist areas, so that they do not dominate the metric. The quality factor Q_{score} reflects customer perception and is calculated as the average of two normalized measures: a Bayesian-adjusted Google Maps rating and a VADER-based sentiment score extracted from review texts [13], as defined by:

$$Q_{score} = \frac{1}{2} \left(\frac{R_{bayes}}{5} + S_{nlp} \right) \quad (2)$$

where R_{bayes} is the Bayesian-adjusted rating, which is normalized to a $[0, 1]$ scale by dividing by 5 within the equation, and $S_{nlp} \in [0, 1]$ is the sentiment score derived from VADER analysis of translated review texts.

The Bayesian adjustment helps reduce bias from locations that have very high ratings but only a small number of reviews. Specifically, the Bayesian-adjusted rating R_{bayes} is computed as:

$$R_{bayes} = \frac{v}{v+m} R + \frac{m}{v+m} C \quad (3)$$

where R is the raw average rating of the establishment, v is its total review count, C represents the global mean rating across all evaluated locations, and m is a confidence threshold defined as the median review count of the dataset.

The factor $E_{neighbor}$ represents the economic strength of nearby areas. Instead of using only the light intensity of the target cell, we compute the average VIIRS NTL value from the surrounding 3×3 grid (excluding the center), denoted as $NTL_{neighbor}$. The multiplier is defined as:

$$E_{neighbor} = 0.5 + \frac{NTL_{neighbor}}{\max(NTL)} \quad (4)$$

where $\max(NTL)$ represents the maximum Night-Time Light intensity value observed across the entire study area. This formulation ensures a baseline value of 0.5, which increases in areas with higher levels of economic activity.

To validate this heuristic before assigning class labels, we examined the relationship between each input signal and the final label. Spearman correlation results showed significant positive relationships for all variables (Cafe_Count: $\rho=0.632$; Total_Reviews: $\rho=0.634$; Avg_Weighted_Rating: $\rho=0.630$; NTL_Neighbor: $\rho=0.679$; all $p<0.0001$). These results support the use of V_{raw} as a proxy for commercial potential.

Although V_{raw} provides a continuous score, classification models require discrete labels. Therefore, we define a rule-based labeling scheme to assign each grid cell into one of three classes ($Y_i \in 0,1,2$), based on both current performance and potential opportunities:

- Class 2 (High Potential): Cells that already contain Coffee & Tea businesses and have a V_{raw} value above the global median of all non-zero scores.
- Class 1 (Medium Potential): This class includes two cases. First, cells with existing businesses but with V_{raw} below the median, indicating moderate performance. Second, empty cells located in economically strong areas. If the $NTL_{neighbor}$ of an empty cell is above the 85th percentile among all empty cells, it is also assigned to Class 1.
- Class 0 (Low Potential): All remaining empty cells that do not meet the above conditions.

2.5 Proposed Architecture

To effectively map the extracted multimodal features to commercial potential labels, we use a customized interaction-aware architecture inspired by the DeepStore framework [3], building on the Wide & Deep paradigm [6] and Deep & Cross Network design [7]. Although our model is designed for a data-scarce setting, it retains the dual-network design of the original approach to capture both explicit and implicit feature interactions.

As illustrated in Fig 2., the proposed architecture includes four main components: an embedding layer, a cross network, a deep network, and a combination layer. Sparse categorical features are embedded into dense vectors, while 3D spatial tensors are processed via a channel-spatial attention mechanism [14] to capture spatial dependencies

across neighboring cells. The resulting unified representation x_0 is passed into two parallel learning paths.

The first path is the cross network, which explicitly models higher-order feature interactions within a limited depth. As shown in the architecture, this component uses a hybrid interaction strategy: embedded sparse and spatial features interact at the vector level to preserve their latent structures. The feature crossing operation at the l -th layer is defined as:

$$x_{l+1} = x_0 x_l^T w_l + b_l + x_l \quad (5)$$

where x_l and x_{l+1} are the outputs of the l -th and $(l + 1)$ -th cross layers, respectively, and w_l and b_l are learnable weight and bias parameters.

Operating in parallel, the deep network pathway is implemented as a standard fully connected Multi-Layer Perceptron (MLP), where the forward propagation at layer l is defined as $h_l = \sigma(W_l^d h_{l-1} + b_l^d)$ with learnable weights W_l^d and biases b_l^d . Finally, to balance memorization and generalization, the combination layer concatenates the output vectors from the linear part, the cross network, and the deep network. The final merged representation is passed through a standard s -class softmax classifier ($s = 3$) to predict class probabilities. The entire network is trained end-to-end by minimizing the standard categorical cross-entropy loss with an added L_2 regularization term to mitigate overfitting.

3 Experimental Setup

3.1 Dataset Preparation

To establish a robust foundation for our multimodal architecture, we first collect data from three major cities in Vietnam - Da Nang, Hanoi, and Ho Chi Minh City - which represent diverse urban structures and economic conditions. Based on this, we construct a comprehensive geospatial dataset consisting of 73,974 valid urban grid cells (with a resolution of $250\text{m} \times 250\text{m}$). The raw data extraction process includes 8,647 Coffee & Tea venues and 51,727 functional Point-of-Interest (POI) entities, which are carefully filtered to remove non-commercial areas such as forests and water bodies.

These spatial entities, together with macroeconomic Night-time Light (NTL) indicators, serve as the foundation for extracting both 13-dimensional tabular features and 11-channel 3D spatial tensors for each grid cell. To capture the micro-level functional diversity of the urban environment, the extracted POI records are further processed through a multi-level rule-based classification pipeline and grouped into eight core functional categories. The final composition of the dataset, detailing both the overarching spatial statistics and the fine-grained POI distribution across the three cities, is summarized in Fig 3. and Table 1.

Table 1. Dataset statistics per city after spatial filtering.

City	Grid cells	Coffee & Tea venues	Total POIs
------	------------	---------------------	------------

Ha Noi	41,341	3,685	22,143
Ho Chi Minh City	27,886	4,081	20,694
Da Nang	4,747	881	8,890
Total	73,974	8,647	51,727

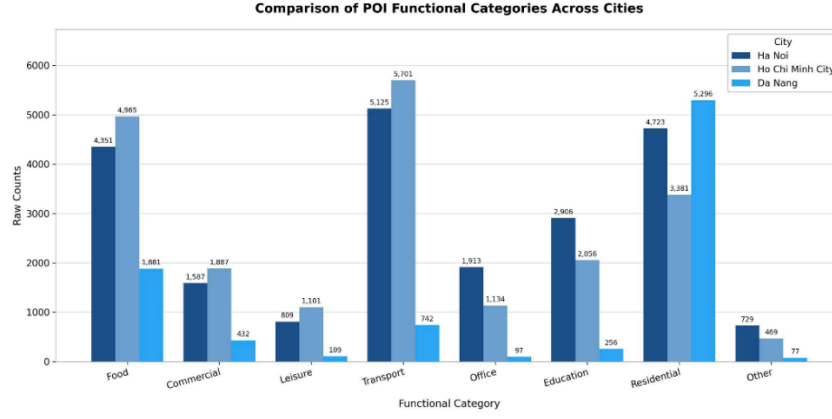


Fig. 3. Comparison of POI functional categories and their raw counts across Ha Noi, Ho Chi Minh City, and Da Nang.

To illustrate how these features are spatially encoded, Fig 4. presents the 11-channel spatial tensor of a representative High-potential grid cell (Class 2). Each panel visualises the 11×11 neighbourhood window centred on the target cell (highlighted by the cyan box). The POI Transport channels show dense activity in surrounding cells, while NTL_{Mean} exhibits strong radiance across the neighbourhood - collectively signalling a commercially active zone. POI Density and POI Entropy further capture the aggregate intensity and functional diversity of land use, providing the spatial attention mechanism with rich contextual signals that a simple tabular feature vector cannot encode.

In the absence of proprietary revenue data, a heuristic labeling approach is used to assign each grid cell to one of three commercial potential levels: Class 0 (Low), Class 1 (Medium), and Class 2 (High). The initial labeling results show a strong class imbalance, which is typical in urban commercial settings. Class 0 dominates with 57,154 cells (77.3%), followed by Class 1 with 13,454 cells (18.2%), while Class 2 - the most commercially active areas - includes only 3,366 cells (4.6%). To assess the model's ability to generalize across different urban structures, the dataset is split using a city-stratified approach. Specifically, 80% of the grid cells from each city are used for training, while the remaining 20% are reserved as a test set, resulting in 14,797 samples. From the training portion, an additional 20% is set aside as a validation set, allowing for careful hyperparameter tuning and evaluation without data leakage.

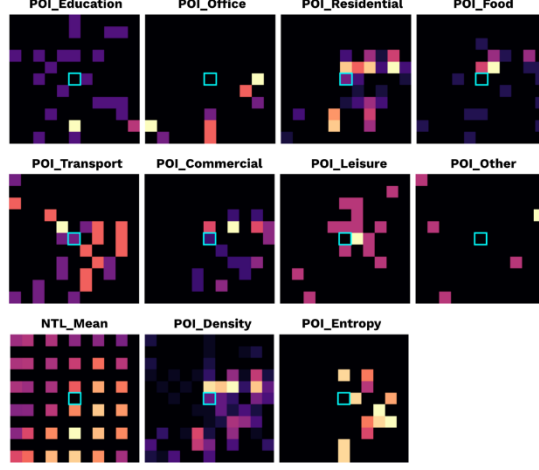


Fig. 4. Spatial tensor visualisation of a representative High-potential grid cell (Class 2).

To address the severe class imbalance (77.3% Class 0, 18.2% Class 1, and 4.6% Class 2), we adopt a hybrid balancing strategy [15]. First, random undersampling is applied to the majority class (Class 0) in the training set, restricting its size to three times that of Class 1, resulting in 21,237 Class 0, 7,079 Class 1, and 1,628 Class 2 samples. Subsequently, balanced class weights are dynamically computed using an inverse-frequency heuristic (0.4700 for Class 0, 1.4100 for Class 1, and 6.1310 for Class 2) and incorporated into the loss function. This weighting scheme penalizes misclassifications of minority classes, guiding the model toward a more balanced learning process.

3.2 Baseline Models and Implementation

We evaluate our architecture against four baselines. The first is Logistic Regression, which serves as a simple linear classifier, providing a reference for basic performance. The second is a Deep Neural Network (DNN) is implemented using a three-layer fully connected structure with hidden sizes of 128, 64, and 32. The Wide & Deep model [6] extends this setup by combining a linear component with the same deep network, but it processes only tabular features. Importantly, the Deep & Cross Network (DCN) [7] baseline is configured with the same cross-network depth and training setup as the proposed model, but it does not include the spatial attention embedding branch. All models were implemented in TensorFlow/Keras and trained using the Adam optimizer [16] with early stopping, Dropout Layer (0.3) [17], and L2 regularization ($\lambda=10^{-3}$).

3.3 Evaluation Metrics

Due to the severe class imbalance in our spatial dataset (where Class 0 accounts for over 77%), overall Accuracy can be misleading. Therefore, Macro F_1 , which treats all

classes equally regardless of size, is adopted as the primary evaluation metric [18], while Weighted F_1 is reported as a complementary measure.

4 Results and Discussion

4.1 Overall Performance Evaluation

As summarized in Table 2, the proposed interaction-aware architecture outperforms all baseline models across every metric. Notably, it delivers a substantial improvement in Macro F_1 over the linear Logistic Regression baseline and neural architectures like Wide & Deep and DCN. The relatively strong performance of Logistic Regression over standard neural baselines is consistent with prior studies on tabular learning under severe class imbalance, where complex models risk overfitting the dominant class. To examine whether ensembling could narrow this gap, we also evaluated a Bagging ensemble of Logistic Regression models. Although Bagging improved the Macro F1 score to 0.7535, it remained below the proposed model’s score of 0.7613. This result suggests that the improvement is not solely due to variance reduction through ensembling, but may also arise from the model’s ability to capture spatial-tabular interactions. The proposed model successfully overcomes this limitation via its multimodal spatial processing component. The performance gap between our model and the DCN baseline highlights the benefit of the two-stage channel-spatial attention mechanism that captures rich neighborhood context lost during text flattening.

Table 2. Performance comparison on the test set.

Model	Accuracy	F1 Macro	F1 Weighted
Logistic Regression	0.9165	0.7073	0.9219
Bagging (LR base)	0.9303	0.7535	0.9337
DNN Baseline	0.8699	0.6215	0.8865
Wide & Deep	0.8667	0.6167	0.8844
DCN	0.8700	0.6364	0.8873
Proposed Model	0.9449	0.7613	0.9489

To evaluate practical utility, the full model’s predictions are visualized in Fig 5. High-potential zones (red) align precisely with major commercial corridors and transportation hubs, transitioning logically into medium-potential areas (orange). This spatial distribution validates the framework’s capability to capture real-world urban commercial structures.

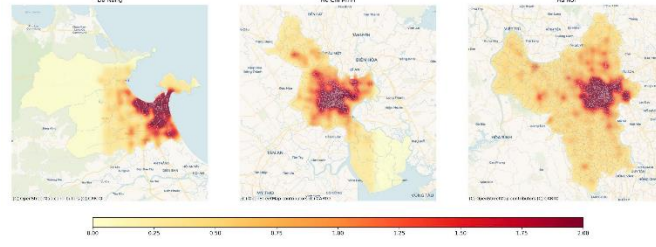


Fig. 5. Predicted site potential heatmaps for Da Nang (left), Ho Chi Minh City (centre), and Ha Noi (right).

4.2 Ablation Study

To evaluate the contribution of each data modality and architectural component, we conduct an ablation study by training each variant from scratch with a fixed random seed (42) for reproducibility. Table 3 presents the performance metrics and the relative change (Δ Macro F1) compared to the full model.

Table 3. Ablation study results on the held-out test set. Δ denotes change in Macro F1 relative to the full model.

Model	Accuracy	F1 Macro	Δ F1 Macro	Training time (s)
Full model	0.9449	0.7613	-	73.3
Only Tabular data	0.8744	0.6400	-0.1213	70.3
Only Spatial data	0.9230	0.6897	-0.0716	46.8
w/o Channel attention	0.9346	0.7309	-0.0304	43.5
w/o Spatial attention	0.8932	0.6494	-0.1119	68.7
w/o Cross Network	0.9467	0.7563	-0.0050	66.1
w/o Deep Network	0.9305	0.7223	-0.0390	45.9
w/o Wide Branch	0.9351	0.7232	-0.0381	47.9

The ablation results highlight that spatial processing components are the primary drivers of the model's predictive performance. Notably, removing the spatial attention module triggers the most severe performance degradation, with a Macro F1 drop of $\Delta = -0.1119$. This underscores the indispensability of dynamically weighting adjacent urban structures rather than treating neighboring signals uniformly. Relying on a single modality causes even larger declines - $\Delta = -0.1213$ when only tabular features are used, and $\Delta = -0.0716$ when only the spatial tensor is retained - confirming that spatial and tabular signals are complementary rather than redundant.

In contrast, the remaining architectural components contribute more moderate, comparable effects. Removing the deep network (MLP) and the wide branch leads to declines of $\Delta = -0.0390$ and $\Delta = -0.0381$, respectively, reflecting their complementary roles in balancing memorization and generalization. The channel attention module yields a smaller, though still non-negligible, contribution ($\Delta = -0.0304$), suggesting that channel-wise reweighting of the 11 input geospatial layers acts as a secondary refinement on top of the dominant spatial attention mechanism. Excluding the cross network yields the smallest drop ($\Delta = -0.0050$), indicating that explicit higher-order feature crossing plays only a minor role in this data-scarce setting compared to spatial contextualization.

5 Conclusion

This study introduces an open-source geospatial proxy-label framework for Coffee & Tea site selection in data-scarce emerging markets. Evaluated on 14,797 grid cells across three Vietnamese cities, our multimodal model achieves a Macro F_1 score of 0.7613, outperforming all baselines, and a high recall of 0.84 for high-potential zones. Ablation studies confirm that the spatial attention mechanism is the most critical component, proving the necessity of spatially weighted context over flattened tabular representations.

However, several limitations remain. The proxy-labeling method depends on existing business density, creating a circular effect where Class 2 is restricted to locations with active businesses, preventing vacant cells from being labeled as high-potential. Additionally, model performance is evaluated against proxy-derived labels rather than empirical revenue or survival rates; thus, its practical predictive validity requires future field validation. Finally, reliance on OpenStreetMap introduces data gaps in rapidly developing peri-urban areas, which macro-level Night-Time Light data cannot entirely resolve. Moreover, because the experiments were conducted only in Vietnamese cities, the relationships learned from open geospatial data may not generalize to markets with different urban structures, POI distributions, or consumer behaviors. Therefore, cross-country validation is needed before broader application.

Future work will focus on four areas as refining sentiment analysis with multilingual language models, integrating alternative open data like commercial rent or mobility patterns to mitigate the dependency on existing business presence, and transitioning from a static snapshot to longitudinal time-based analysis to capture dynamic urban evolution, and extending the framework to cities beyond Vietnam to assess cross-market generalizability.

References

1. Ferreira, J., Ferreira, C.: Challenges and opportunities of new retail horizons in emerging markets: The case of a rising coffee culture in China. *Business Horizons* 61(5), 783–796 (2018)

2. Wu, M., Pei, T., Wang, W., et al.: Roles of locational factors in the rise and fall of restaurants: A case study of Beijing with POI data. *Cities* 113, 103185 (2021)
3. Liu, Y., Yao, L., Guo, B., et al.: DeepStore: An interaction-aware wide & deep model for store site recommendation with attentional spatial embeddings. *IEEE Internet of Things Journal* 6(4), 7319–7333 (2019)
4. Yan, H., Wang, S., Yang, Y., et al.: O2-SiteRec: Store site recommendation under the O2O model via multi-graph attention networks. In: 2022 IEEE 38th International Conference on Data Engineering (ICDE). IEEE (2022)
5. Ting, C.-Y., Ho, C.C., Yee, H.J., et al.: Geospatial analytics in retail site selection and sales prediction. *Big Data* 6(1), 42–52 (2018)
6. Cheng, H.-T., Koc, L., Harmsen, J., et al.: Wide & deep learning for recommender systems. In: *Proceedings of the 1st Workshop on Deep Learning for Recommender Systems*, Boston, MA, USA. ACM (2016)
7. Wang, R., Fu, B., Fu, G., et al.: Deep & cross network for ad click predictions. In: *Proceedings of ADKDD'17*, Halifax, NS, Canada. ACM (2017)
8. Wang, L., Fan, H., Wang, Y.: Site selection of retail shops based on spatial accessibility and hybrid BP neural network. *ISPRS International Journal of Geo-Information* 7(6), 202 (2018)
9. Haklay, M.: How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets. *Environment and Planning B: Planning and Design* 37(4), 682–703 (2010)
10. Suleiman, H., Nguyen, M.-T.T., Mendez, C.: Predicting subnational GDP in Vietnam with remote sensing data: A machine learning approach. *Letters in Spatial and Resource Sciences* 18(1), 5 (2025)
11. Elvidge, C.D., Baugh, K., Zhizhin, M., et al.: VIIRS night-time lights. *International Journal of Remote Sensing* 38(21), 5860–5879 (2017)
12. Xu, Y., Cong, G., Zhu, L., et al.: MMPOI: A multi-modal content-aware framework for POI recommendations. In: *Proceedings of the ACM Web Conference 2024*, Singapore. ACM (2024)
13. Hutto, C., Gilbert, E.: VADER: A parsimonious rule-based model for sentiment analysis of social media text. In: *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, vol. 8, no. 1, pp. 216–225 (2014)
14. Woo, S., Park, J., Lee, J.-Y., et al.: CBAM: Convolutional Block Attention Module. In: Ferrari, V., Hebert, M., Sminchisescu, C., et al. (eds.) *Computer Vision – ECCV 2018*. LNCS, vol. 11211, pp. 3–19. Springer, Cham (2018)
15. He, H., Garcia, E.A.: Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering* 21(9), 1263–1284 (2009)
16. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. arXiv:1412.6980 (2014)
17. Srivastava, N., Hinton, G., Krizhevsky, A., et al.: Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research* 15(56), 1929–1958 (2014)
18. Powers, D.M.: Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. arXiv:2010.16061 (2020)